

Efficient 3D Reconstruction of NeRF using Camera Pose Interpolation and Photometric Bundle Adjustment

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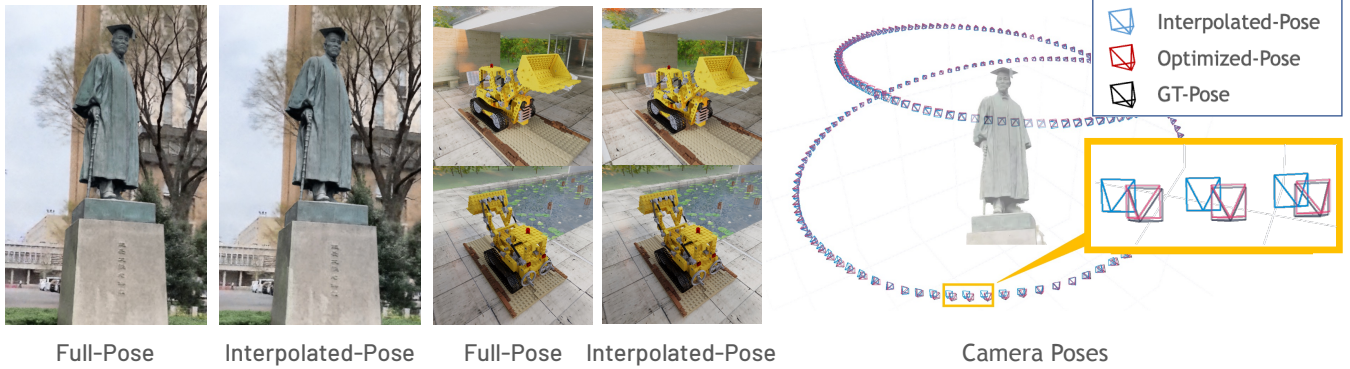


Figure 1: Comparison of the results of learning NeRF with the full camera poses and the interpolated-poses. The Full-Pose result uses the camera poses obtained by COLMAP with all the input images. The Interpolated-Pose result uses the poses obtained by COLMAP with several images and the interpolated poses between them as the initial poses. We apply Catmull-Rom Spline interpolation for translations and SLERP interpolation for rotations. The figure on the right shows the visualization of the camera poses using synthetic data. Interpolated-Pose generates images with almost the same quality as Full Pose.

CCS CONCEPTS

• **Computing methodologies** → **Camera calibration; Reconstruction; Computer systems organization** → **Embedded systems; Redundancy; Robotics**; • **Networks** → **Network reliability.**

KEYWORDS

Novel View Synthesis, Neural Radiance Fields, Camera Calibration

ACM Reference Format:

Tsukasa Takeda, Shugo Yamaguchi, Kazuhito Sato, Kosuke Fukazawa, and Shigeo Morishima. 2023. Efficient 3D Reconstruction of NeRF using Camera Pose Interpolation and Photometric Bundle Adjustment. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters*

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SIGGRAPH '23 Posters, August 06–10, 2023, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-0152-8/23/08.

<https://doi.org/10.1145/3588028.3603691>

(SIGGRAPH '23 Posters), August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588028.3603691>

1 INTRODUCTION

Recently, there has been a surge of interest in Neural Radiance Fields (NeRF) [Mildenhall et al. 2020] for synthesizing novel views from images captured from different viewpoints and camera poses. NeRF is capable of high-definition view synthesis by leveraging multilayer perceptrons (MLPs) to map the color and density to every point in the 3D space. However, a major drawback of NeRF is its requirement for camera parameters (intrinsics and extrinsics). COLMAP [Schonberger and Frahm 2016] can estimate poses by matching Scale Invariant Feature Transform (SIFT) feature points, which is computationally intensive, especially when there are many images. Despite the drawback that limits the practical application of NeRF, few studies have focused on this issue.

In this study, we propose a novel approach that speed up 3D reconstruction of NeRF by interpolating camera poses. First, we select low-blurred images from adjacent frames and sparsely estimate the camera poses using COLMAP, assuming sequential data captured by a monocular camera. Then, we estimate the camera

Table 1: Computation time of COLMAP

scene	input images	GPU times
Statue	340	46m05s
	34	0m49s
Doll	370	77m26s
	37	0m32s

poses for the unposed images by interpolating the poses from neighboring estimated poses. Finally, we train NeRF with Photometric Bundle Adjustment [Lin et al. 2021] to optimize the camera pose and Radiance Field simultaneously using all available images. Our experimental results using both real and synthetic data show that this approach significantly reduces the time required for camera pose estimation with only a slight decrease in quality.

2 OUR APPROACH

Our goal is to perform 3D scene reconstruction for NeRF including camera pose estimation in a fast and efficient manner. We propose the following three-step approach.

- (1) Select frames and estimate camera poses.
- (2) Interpolate camera poses.
- (3) Learn NeRF with Photometric Bundle Adjustment.

First, we select low blurred images from the captured adjacent frames and use COLMAP for pose estimation. We compute the image score of the Laplacian and select the images with the highest score within a certain interval. Second, we interpolate unposed cameras by estimated camera poses. Camera poses are represented by translations and rotations. For translations, we conducted two interpolation methods: linear interpolation and Catmull-Rom Spline interpolation which creates a smooth curve that passes through adjacent points. For the rotational component, we applied SLERP (spherical linear interpolation), which interpolates between two vectors of equal length on a circular arc. Finally, using the camera poses estimated by COLMAP and poses generated by aforementioned interpolation, we optimize the camera poses $\mathbf{p}_i$ simultaneously with the radiance field Θ with Photometric Bundle Adjustment using all images.

$$\min_{\mathbf{p}_1, \dots, \mathbf{p}_M, \Theta} \sum_{i=1}^M \sum_{\mathbf{u}} \left\| \hat{\mathcal{I}}(\mathbf{u}; \mathbf{p}_i, \Theta) - \mathcal{I}_i(\mathbf{u}) \right\|_2^2 \quad (1)$$

where $\mathbf{u} \in \mathbb{R}^2$ represents given pixel coordinates, $\hat{\mathcal{I}}$ represents rendered image and $\{\mathcal{I}_i\}_{i=1}^M$ represents input M images. Unlike COLMAP, which estimates pose based on feature points, only a small computational cost is required because pose can be optimized using only photometric loss shown in Eq.(1).

3 EXPERIMENTS

To validate the effectiveness of our approach, we conducted separate experiments on real and synthetic data.

First, we conducted experiments on real data to validate how the method can shorten pose estimation. The real data uses images captured with the 360 degree object-centric trajectory and sparsifying the input image by about 1/10. Table 1 shows the time required for pose estimation using the images of the two scenes as input to

Table 2: Quantitative results

posed/train images	translational interpolation	PSNR↑	SSIM↑	LPIPS↓
360/360	off	28.20	0.85	0.10
36/36	off	12.78	0.41	0.61
36/360	Catmull-Rom	27.72	0.84	0.11
36/360	Linear	21.46	0.61	0.20

COLMAP. Based on these results, the time required for pose estimation was successfully accelerated by about 100 times by reducing the input image to COLMAP by about 1/10. The image quality of the statue scene is shown in Fig 1. Please see Supplementary Materials for other rendering results.

Next, we conducted experiments on Blender synthetic dataset [Ma et al. 2021] with ground-truth images in test data to validate the quality of the generated images. We evaluated the score of PSNR, SSIM and LPIPS for the four case classifications shown in Table 2. Table 2 shows that the quality of rendered images using interpolated poses is lower than using ground-truth poses when all images are given. However, it is able to generate higher quality images by utilizing interpolated poses to train NeRF, even when using unposed images. Then, we found that Catmull-Rom Spline interpolation in the translations closer to the real camera motion allows for higher quality scene reconstruction.

Our experiments were conducted based on the NerfStudio project (nerfacto models) [Tancik et al. 2023] and a single Nvidia RTX 2080Ti GPU.

4 CONCLUSION AND DISCUSSIONS

This paper attempts to use camera pose interpolation and Photometric Bundle Adjustment for efficient scene reconstruction with NeRF. These experiments show that the time required for pose estimation was greatly accelerated with only a slight decrease in quality. However, the number of image reduction of our approach is limited because COLMAP requires a certain minimum number of images to reconstruct. We consider to build a network that can automatically estimate the most efficient image selection to balance the computation time and the reconstruction quality.

ACKNOWLEDGMENTS

This work was supported by JSPS KAKENHI Grant Numbers JP21H05054.

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